案例：专家学习与机器学习

机器翻译自动评价方法：BLEU

杨沐昀
语言技术研究中心

Motivation

- Why automatic evaluation for MT?
  - Manual evaluation is expensive, inconsistent and time consuming.
  - MT development need instant feedback on his efforts
    - Whether my algorithm, my model, new weight help?
  - Large scale, objective evaluation is of substantial significance for any research.

How ?

- Do we need to study how people recognize good translation?
  - Word, phrase, sentence structure and pattern?
  - A long history of translation argues what is good translation!
- In most cases, “whether better” matters more than “how better”!
- Can we accomplish this by a simple way?

Observations!

- The closer a (machine) translation is to a professional human translation, the better it is!
  - A corpus of good quality human reference translations
  - A numerical translation closeness metric!!!

Examples

Example 1:

Candidate 1: It is a guide to action which ensures that the military always obeys the commands of the party

Candidate 2: It is to insure the troops forever hearing the activity guidebook that party direct

Reference 1: It is a guide to action that ensures that the military will forever heed party commands

Reference 2: It is the guiding principle which guarantees the military forces always being under the command of the party

Reference 3: It is the practical guide for the army always to heed the directions of the party

Match Counting?

- Ranking the candidates
  - Simply comparing the candidate translation and the reference translations and counts the number of matches.
- Assumption 1: simple counting method (by unigram word)
  - Counting the number of candidate translation words which occur in any reference translation and then divides by the total number of words in the candidate translation

Exhausted Counting

Example 2：

- Candidate: *the the the the the the the*
- Reference1: *the cat is on the mat.*
- Reference2: *there is a cat on the mat.*

  - Simple standard unigram count is 7/7;
  - Each word should be modified as exhausted after the match identified;
  - Thus, the modified unigram precision is *2/7*;

Modified Bigram Precision

Example 1:

Candidate 1: It is a guide to action which ensures that the military always obeys the command of the party

Candidate 2: It is to insure the troops forever hearing the activity guidebook that party direct

Reference 1: It is a guide to action that ensures that the military will forever heed party commands

Reference 2: It is the guiding principle which guarantees the military forces always being under the command of the party

Reference 3: It is the practical guide for the army to heed the directions of the party

- candidate 1 achieves a modified bi-gram precision of 10/17
- whereas the candidate 2 achieves a modified precision of 1/13.

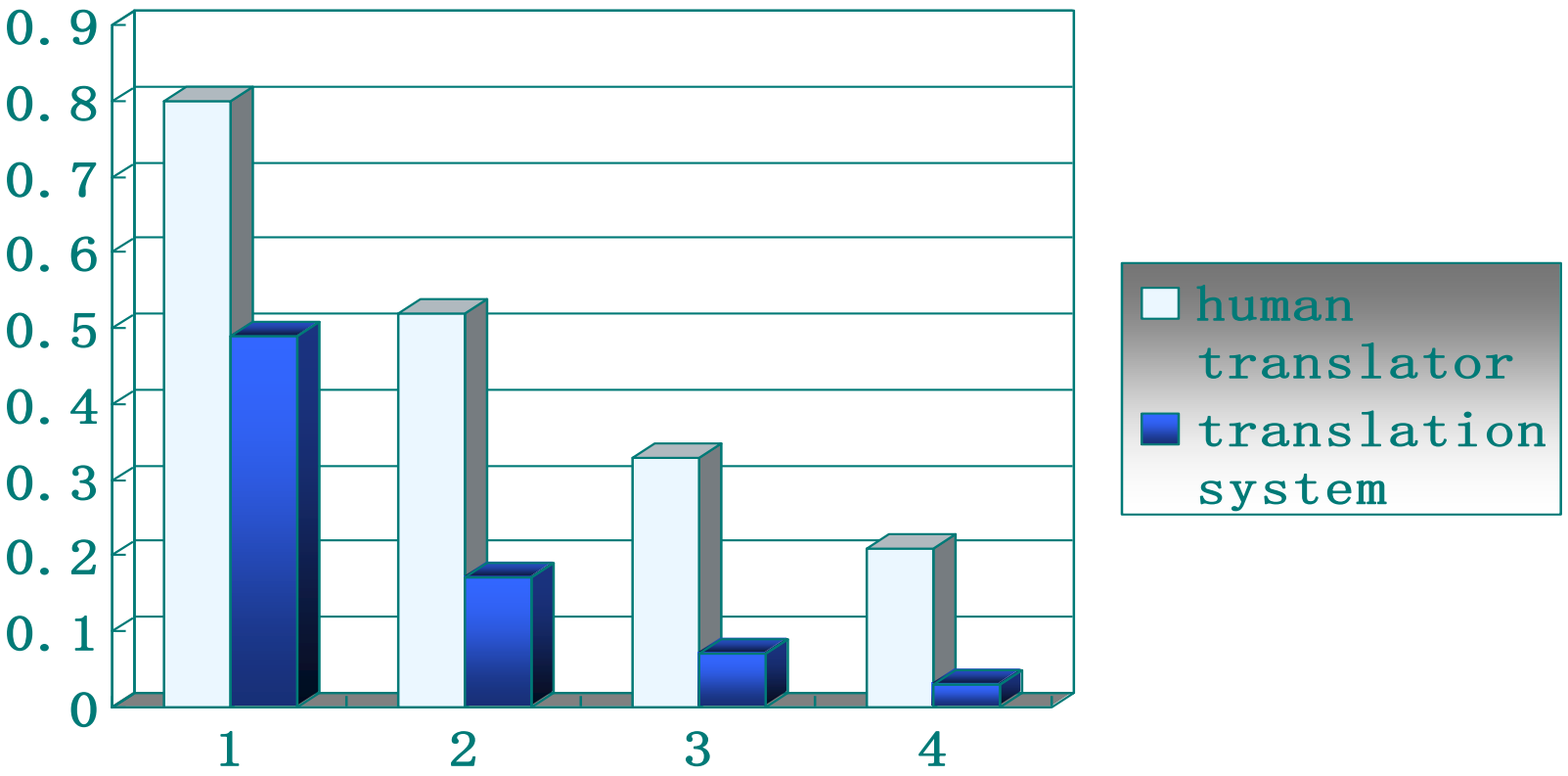
Are We Reasonable

- This sort of modified n-gram precision scoring captures two aspects of translation quality
  - Unigram tends to satisfy adequacy (忠实度)
  - The longer n-gram matches account for fluency (流利度);

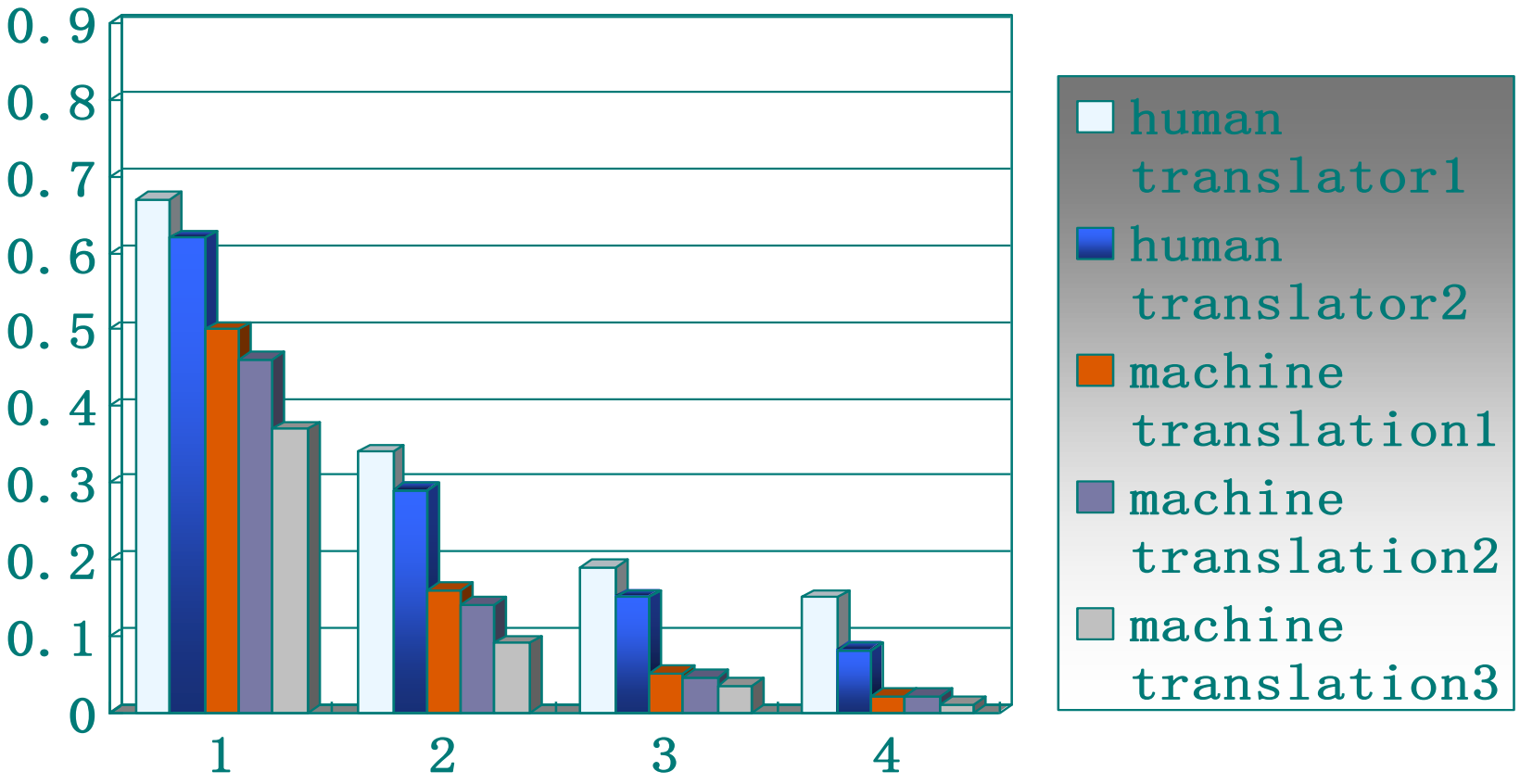
Modified n-gram Precision on Translation Text

$$Pn = \frac{\sum_{C \in \{candidates\}} \sum_{n-gram \in C \,\&\, Ref} Count_{clip}(n-gram)}{\sum_{C \in \{candidates\}} \sum_{n-gram \in C} Count(n-gram)}$$

Compare Human Translator and a Translation System



Compare Multiple Human Translators and MT Systems



How to Combine

- Average?
- Note the modified *n*-gram precision decays roughly exponentially with *n*:
  - Unigram>Bi-gram >> trigram
- How to take account of this?
  - Smooth the sharp difference in average!

Problem: Sentence Length

- Recall Issue

  Candidate1:of the

  Reference1: It is a guide to action that ensures that the military will forever heed party commands

  Reference2: It is the guiding principle which guarantees the military forces always being under the command of the party

  Reference3: It is the practical guide for the army to heed the directions of the party

- The modified unigram precision is 2/2, and the modified bigram precision is 1/1!

Solution from Mathematics

- We may penalize the short ones with a brevity penalty;
  - （Precision may balance long sentences）
- Average logarithm against arithmetic average and geometric mean?
  - Log is a good smoothing function!

BLEU Metric

$$BLEU = BP \bullet \exp\left(\sum_{1}^{N} w_n \log p_n\right)$$

$$BP = \begin{cases} 1 & if\ c > r \\ e^{(1-r/c)} & if\ c \le r \end{cases}$$

$$\log BLEU = \min(1 - \frac{r}{c}, 0) + \sum_{1}^{N} w_n \log p_n$$

$$N = 4, w_n = 1/N$$

BLEU: An Example

- **Candidate 1: the book is on the desk**

- **Ref1: there is a book on the desk**
- **Ref2: the book is on the table**

unigram:	bigram:	trigram:
	$Count_{clip}(the, book) = 1$	$Count_{clip}(the, book, is) = 1$
	$Count_{clip}(book, is) = 1$	$Count_{clip}(book, is, on) = 1$
	$Count_{clip}(is, on) = 1$	$Count_{clip}(is, on, the) = 1$
	$Count_{clip}(on, the) = 1$	$Count_{clip}(on, the, desk) = 1$
	$Count_{clip}(the, desk) = 1$	
$\sum_{unigram \in C} Count(unigram) = 6$	$\sum_{bigram \in C} Count(bigram) = 5$	$\sum_{trigram \in C} Count(trigram) = 4$
$p_1 = 1$	$p_2 = 1$	$p_3 = 1$

$$\left.\begin{array}{l} c = 6 \\ r = 6 \end{array}\right\} = e^{1-\frac{r}{c}} = e^{0} = 1 = BP$$

$$BLEU = BP \bullet \exp\left(\sum_{n=1}^{N} w_n \log p_n\right)$$

$$= \exp\left[\frac{1}{3}(\log 1 + \log 1 + \log 1)\right] = 1$$

BLEU Evaluation--Consistency

Figure 5: BLEU predicts Monolingual Judgments

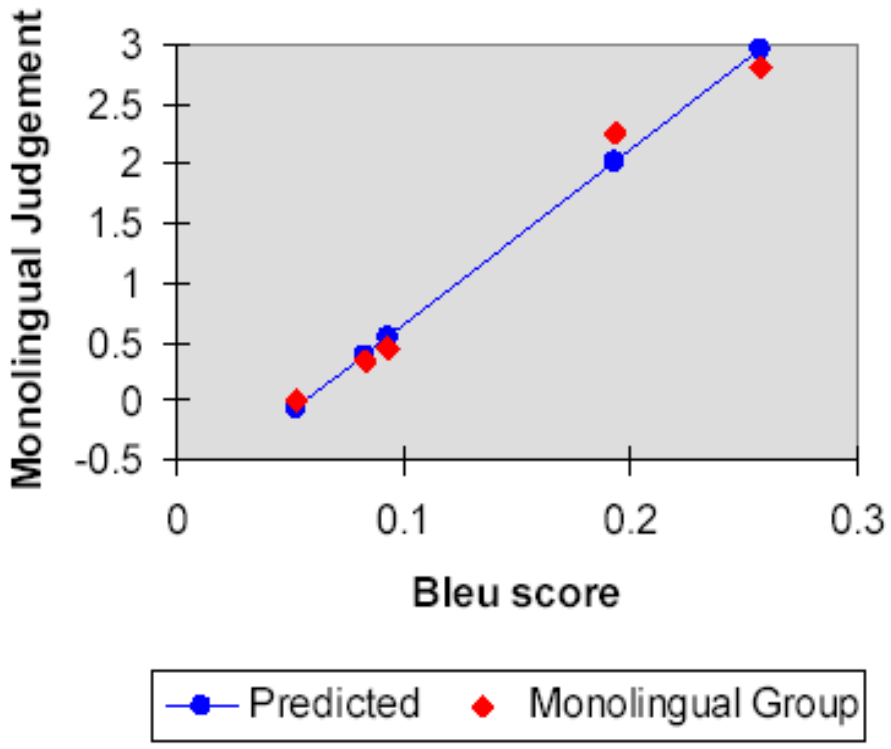
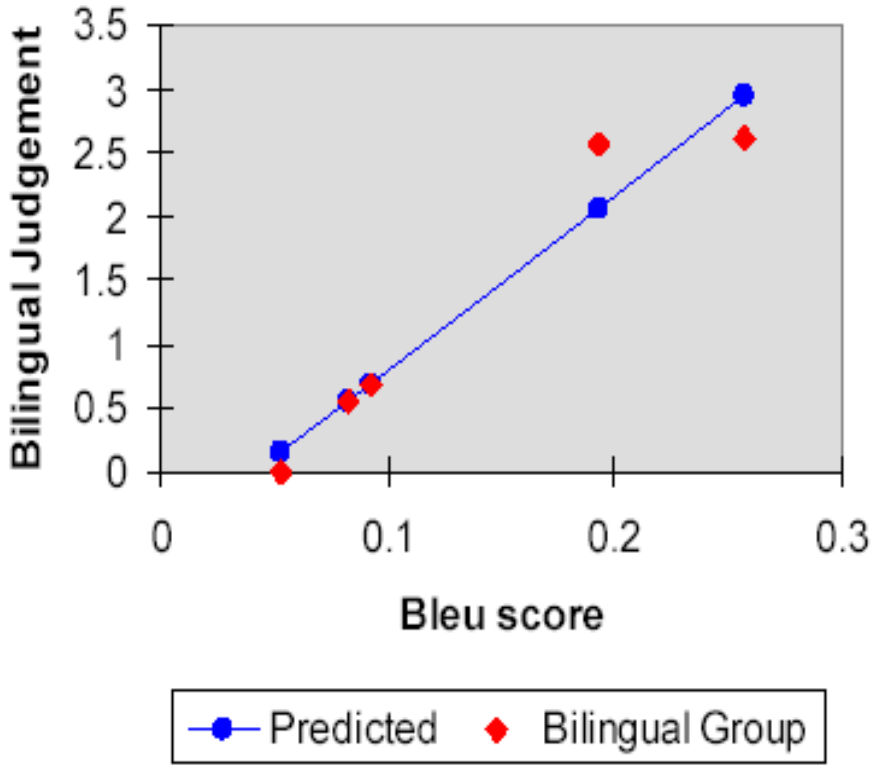


Figure 6: BLEU predicts Bilingual Judgments



Adopted by NIST for TIDES Project

- Corpus used to evaluation of N-gram Scoring

Corpus	Source language	#of documents	#of human translations	#MT systems
DARPA 1994 French-English	French	100	2	5
DARPA 1994 Japanese-English	Japanese	100	2	4
DARPA 1994 Spanish-English	Spanish	100	2	4
DARPA 2001 Chinese-English	Chinese	80	11	6

Correlation between BLEU Score and Human Assessment

Corpus	Systems	Adequacy (%)	Fluency (%) s	Informatics (%)
DARPA 1994 French-English	5 MT systems	95.7	99.7	91.4
DARPA 1994 Japanese-English	4 MT systems	97.8	85.6	98.3
DARPA 1994 Spanish-English	4 MT systems	97.5	97.2	94.3
DARPA 2001 Chinese-English	6 Commercial systems	95.2	97.1	-

Typical BLEU scores (2005 NIST evaluation data)

Arabic-English news translation, 4 references:

- Best statistical (research) system: 51% BLEU score
- (some) commercial systems: 10 - 34% BLEU score
- Estimated human BLEU score: 63% BLEU score (see next slide)

Chinese-English news translation, 4 references:

- Best statistical (research) system: 35% BLEU score
- (some) commercial system: 15% BLEU score
- Estimated human BLEU score: 55% BLEU score (see next slide)

References

- The website for NIST MT Evalution: http://www.nist.gov/speech/tests/mt/index.htm
- *BlEU: a method for automatic evaluation of machine translation,* Kishore Papieni, Salim Roukos, Todd Ward, Wei-Jing Zhu, ACL 2002.

第5讲专题

BLEU：专家学习到机器学习下篇

杨沐昀

哈工大教育部-微软语言语音重点实验室

MOE-MS Joint Key Lab of NLP and Speech (HIT)

Where Could BLEU Be Wrong

$$BLEU = BP \times \exp\left(\sum_{n=1}^{N} w_n \log p_n\right)$$

- Parameter?
- Equation?
- Algorithm?
- …
- …
- How to reveal it?

Defects in BLEU

- Could BLEU Be Wrong?
  - Challenge it by our common sense
  - How about MT and Human Translation
    - which is better?
    - Evaluate human translation?
- Compare Human *vs* MT Translation quality
  - IWSTL2004：506句，每句16个人工标准译文
  - NIST_mt04 :1788句，每句4个人工标准译文
  - 在标准译文中任意选出一个句子作为候选译文，其余15个作为参考译文，使用BLEU工具进行自动评价
  - Guess the Max and Min score!

Defects in BLEU

- MT Beats Human?

BLEU-4	MaxHuman	MinHuman	Total MTsys	MT > Human
IWSLT2004	0.9704	0.2866	20	13
NIST_mt04	0.5064	0.2824	17	2

Defects in BLEU

- To Evaluate Human Translation
  - 数据： 152篇翻译，阅卷点正式评分
    - 某英语水平考试英汉翻译试题: 12分
    - 1段英文、3个句子
    - 1个标准译文+3个手工译文
  - Word for Chinese, segmentation or not?

各分数段的样本分布

分数	1	2	3	4	5	6	7	8	9	10	11	12
样本数	9	8	7	6	13	14	16	19	25	16	12	7

We offer a wide range of information on the website,	我们在网站上提供大量商品信息，
from detailed information	从详细介绍
to general buying guides in categories(种类)	到一般导购，涉及的商品种类
such as digital cameras, home theaters, clothes and watches.	有数码相机、家庭影院、服装和钟表等。
Whether you make your purchase online	无论你是在网上购物，
or print out the information to take into any of our chain stores,	还是将购物信息打印出来带到我们任何一家连锁商店去订购，
we want you to feel confident	我们希望您对自己做出的购物决定充满信心。
in your purchase decision.	
If one item you order	如订购的某款商品
does not meet your expectations,	达不到你的要求，
simply return it to our store	只管退还给我们的商店，
at no cost.	不受任何损失。

Defects in BLEU

- Performance on Human Translations
  - 4个参考译文时与人工评价的相关性较好；
  - 按字计算时与人工评价的相关性较好；
  - 无论是按字匹配，按词匹配，按词性匹配，还是按词与词性同时匹配的， BLEU的性能<0.7

参考译文数	按字	按词匹配	按词性	词+词性
1	0.573	0.539	0.560	0.548
4	0.684	0.624	0.673	0.620

Defects in BLEU

- Analysis
  - Treat word equally
  - Treat n-gram equally
  - Geometric Mean
  - Ngram is not for structure
  - ……
- Solutions?

NIST Metric: A Weighted Ngram

- Background
  - July 2001 TIDES PI meeting in Philadelphia, IBM described BLEU
  - Compare MT output with expert translations in word N-grams
  - Elegant in its simplicity, and strong correlation with human judgment.
  - DARPA commissioned NIST to develop an MT evaluation facility based on the IBM work.

NIST Metric: A Weighted Ngram

NIST's Examination: Bad performance for HT

The Corpus	The Systems	Adequacy (%)	Fluency (%)	Informativeness (%)
1994 French Corpus	5 MT Systems	95.7	99.7	91.4
1994 Japanese Corpus	4 MT Systems	97.8	85.6	98.3
1994 Spanish Corpus	4 MT Systems	97.5	97.2	94.3
2001 Chinese Corpus	6 Commercial MT Systems	95.2	97.1	-
	7 Professional Translators	70.5	16.6	-

NIST Metric: A Weighted Ngram

- Motivation: weight more heavily those N-grams that are more informative

$$Info(w_1 \ldots w_n) = \log_2 \left(\frac{\text{the \# of occurrences of } w_1 \ldots w_{n-1}}{\text{the \# of occurrences of } w_1 \ldots w_n} \right)$$

- How informative n-gram is favored by this?
  - similar to IDF;

NIST Metric: A Weighted Ngram

$$Score = \sum_{n=1}^{N} \left\{ \frac{\sum_{\substack{\text{all } w_1 \ldots w_n \\ \text{that co-occur}}} Info(w_1 \ldots w_n)}{\sum_{\substack{\text{all } w_1 \ldots w_n \\ \text{in sys output}}} (1)} \right\} \cdot \exp\left\{ \beta \log^2 \left[\min\left(\frac{L_{sys}}{L_{ref}}, 1 \right) \right] \right\}$$

- β is chosen to make the brevity penalty factor = 0.5 when the # of words in the system output is 2/3rds of the average # of words in the reference translation

NIST Metric: A Weighted Ngram

- **Candidate 1: one book is on the desk**
- **Reference 1: there is a book on the**
- **Reference 2: the book is on the table**

unigram:	bigram:	trigram:
$Info(w_i) = \log_2 \frac{\#\ words\ of\ all\ refs}{\#\ occuring\ in\ all\ refs}$	$Info(w_1 \ldots w_n) = \log_2 \frac{the\ \#\ occurences\ of\ w_1 \ldots w_{n-1}\ in\ all\ refs}{the\ \#\ occurences\ of\ w_1 \ldots w_n\ in\ all\ refs}$	
$Info(one) = null$	$Info(one, book) = null$	$Info(one, book, is) = null$
$Info(book) = \log_2(13/2)$	$Info(book, is) = \log_2(2/1)$	$Info(book, is, on) = \log_2(1/1)$
$Info(is) = \log_2(13/2)$	$Info(is, on) = \log_2(2/1)$	$Info(is, on, the) = \log_2(1/1)$
$Info(on) = \log_2(13/2)$	$Info(on, the) = \log_2(2/2)$	$Info(on, the, desk) = \log_2(2/1)$
$Info(the) = \log_2(13/3)$	$Info(the, desk) = \log_2(3/1)$	
$Info(desk) = \log_2(13/1)$		
$\sum_{all\ unigram in\ candidate}(1) = 6$	$\sum_{all\ bigram in\ candidate}(1) = 5$	$\sum_{all\ trigram in\ candidate}(1) = 4$
$\sum Info(w_i) / \sum_{all uni in\ candidate}(1) \approx 2.32$	$\sum Info(w_i w_{i+1}) / \sum_{all bigram in\ candidate}(1) \approx 0.717$	$\sum Info(w_i w_{i+1} w_{i+2}) / \sum_{all trigram in\ candidate}(1) = 0.25$

NIST Metric: A Weighted Ngram

- Candidate 1: one book is on the desk

- Reference 1: there is a book on the
- Reference 2: the book is on the table

brevity penalty:

$$L_{sys} = 6 \qquad \overline{L}_{ref} = 6.5 \qquad \beta = 4.217$$

$$BP = \exp\left\{\beta \log^2\left[\min\left(\frac{L_{sys}}{\overline{L}_{ref}}, 1\right)\right]\right\} = 0.973$$

Score:

$$N = 3$$

$$score = \sum_{n=1}^{N}\left\{\sum Info(w_1 \ldots w_n) \Big/ \sum_{\substack{\text{all } w_1 \ldots w_n \\ \text{in candidate}}} (1)\right\} \bullet BP$$

$$= (2.32 + 0.717 + 0.25) * 0.973$$

$$\approx 3.199$$

NIST Metric: A Weighted Ngram

- Primary Differences with BLEU
  - Geometric mean versus arithmetic sum
  - Uniform weight versus information weight
  - Selective use of n-gram
    - N=4 for BLEU
    - N=5 for NIST

Skip Ngram: Get Different Kind of Ngram

- **Skip N-gram:** any pair of same length n-gram in sentence order, allowing for arbitrary gaps
  - SNR uses the n-gram pair with arbitrary gap **&** Rouge uses the one gram pair with arbitrary gap

E.g. Skip-Ngrams (N <= 2) in "Australia reopens embassy in Manila" :

"Australia, reopens", "Australia, embassy", "Australia, in", "Australia, manila", "reopens, embassy", "reopens, in", "reopens, manila", "embassy, in", "embassy, manila", "in, manila"

"Australia reopens, embassy in", "Australia reopens, in manila", "reopens embassy, in manila"

SNR: Skip *N*-gram Regression

- Skip N-gram Match
  - Partial match: if only one of the two members is matched
  - Full match: if both members are matched
  - Ordered full match: if both members are matched in their sentence order
- Skip N-gram is calculated in recall
  - Partial match recall: M_p
  - Full match recall: M_f
  - Ordered full match recall: M_o

SNR: Skip *N*-gram Regression

- SN score: straight mean of three skip N-gram recall
  - $SN = (M_p + M_f + M_o)/3$
- SNRegression score
  - Assign weights to M_p, M_f and M_o
  - Resolve the weights by SVM
  - Trained by five-fold cross validation on the data provided by NIST Metrics-MATR 2008 evaluation

Source of Data	LDC2008E43
Genre	Newswire
Number of documents	25
Total number of segments	249
Source Language	Arabic
Number of system translations	8
Number of reference translations	4
Human assessment scores	Score 1-7

SNR: Skip *N*-gram Regression

- SNR series are outstanding in all metrics
- SN series exceed all baselines except METEOR
- Comparing with the Rouge, the most similar metric of this work, all novel metrics performs better
- Regression method improves the performance significantly
- Stem is always helpful in this experiment
- Longer gram is not helpful in some cases.

Criterion	Correlation
METEOR	0.705
ROUGE-4	0.654
ROUGE-9	0.663
ROUGE*	0.655
BLEU	0.609
GTM	0.543
SN-1	0.686
SN-4	0.665
SN-1-Stem	0.689
SN-4- Stem	0.670
SNR-1	0.716
SNR-4	0.741
SNR-1- Stem	0.720
SNR-4- Stem	**0.745**

Pearson on segment level

(The SNR scores are the average of 5-fold validation)

Skip Ngram: Get Different Kind of Ngram

- A SVM Regression Based Skip-Ngram Approach to MT Evaluation
  - Longer skipped gram pair.
  - Multiple statistics of skip-Ngram by full, partial and ordered matching
  - Regression to human assessments using multiple statistics as features

LTE: The Fundamental Unit for Ngram

- Method: BLEU counted in letter
- Original BLEU
  - Candidate: As you wish no problem
  - Reference: No problem as you like
- LTE
  - Candidate: A s y o u w i s h n o p r o b l e m
  - Reference: N o p r o b l e m a s y o u l i k e.

LTE: The Fundamental Unit for Ngram

Experiment on training data

LET	1	2	3	4	5	6	7	8
Pear	.264	.548	.656	.682	.687	**.687**	.684	.678
Spea	.484	.609	.674	.700	.711	**.716**	.717	.715

- Why 6-gram is the best?
  - The average word length is 6.

LTE: The Fundamental Unit for Ngram

Compare with other metrics

Correlation	BLEU-4	NIST-5	Meteor0.6	Rouge*
Pearson	0.605	0.735	0.774	0.685
Spearman	0.608	0.686	0.724	0.683

- LTE
  - Pearson: 0.687
  - Spearman: 0.716

LTE: The Fundamental Unit for Ngram

- Advantages
  - Letter match is another resolution to partial word-match;
  - Captures more about translation adequacy
  - Less sensitive to errors of short words; which may be less important;
- Defects
  - Inherited for BLEU framework

引入机器学习：相关因素庞杂

- Many successful metrics available
  - BLEU/NIST/GTM/Rouge/Meteor……
- How far can we go by combining these metrics?
  - Combining metrics to avoid bias
- Can linguistic knowledge helpful?
  - POS is the key knowledge at word level

机器学习工具包

- 很多：Weka/SciKit-Learn/SVM-light 等
- 训练文件格式基本类似

ID；F1；…Fn；Label

  - ID: 样本编号
  - F1…Fn：特征值
  - Label: 样本标签
    - 分类：0,1
    - 回归：实值
    - 排序：排序位置

weather.arff

Relation: weather

No.	1.outlook Nominal	2.temperature Numeric	3.humidity Numeric	4.windy Nominal	5.play Nominal
1	sunny	85.0	85.0	FALSE	no
2	sunny	80.0	90.0	TRUE	no
3	overcast	83.0	86.0	FALSE	yes
4	rainy	70.0	96.0	FALSE	yes
5	rainy	68.0	80.0	FALSE	yes
6	rainy	65.0	70.0	TRUE	no
7	overcast	64.0	65.0	TRUE	yes
8	sunny	72.0	95.0	FALSE	no
9	sunny	69.0	70.0	FALSE	yes
10	rainy	75.0	80.0	FALSE	yes
11	sunny	75.0	70.0	TRUE	yes
12	overcast	72.0	90.0	TRUE	yes
13	overcast	81.0	75.0	FALSE	yes
14	rainy	71.0	91.0	TRUE	no

建模策略：常见类型

- Machine Learning Approach
  - Classification: ?
  - Regression?
    - better, but for another metric
    - Robust and generalization can be achieved via multiple features
  - Ranking: V
    - SVM Ranking
    - Sensitive but not for robust, diagnostic……

Case Sutdy：SVM-Ranking for MTE

- Feature selection: Best-in or Worst-out
- Selected
  - BLEU1, BLEU2, LetterBLEU
  - Meteor
  - ROUGE-9, recall, case insensitive
- Removed
  - BLEU4, NIST5
  - Rouge in other conditions
  - GTM, WER……

Case Sutdy：SVM-Ranking for MTE

- Linguistic info still beneficial?
- How shall we use POS?
  - Each pos is small in quantity in one sentence.
  - Cluster them
- POS Selected
  - Verb: MD,VB, VBD,VBG,VBN,VBP,VBZ
  - Noun: NN,NNS,NNP,NNPS,FW
  - Other: LS, SYM, punctuation
  - Not used: adj, adv, prep……

Case Sutdy：SVM-Ranking for MTE

- In NIST MART Metrics08
  - 10 Top1 out of 144 indexes;
  - Top 2 in weighted score over multiple reference tracks;
  - Top 8 in weighted score over all tracks;
- In WMT 2011
  - 3 champions out of 4, with the last ranked top2.

Case Sutdy：SVM-Ranking for MTE

- Why POS improves performance
  - POS carries word syntax information;
  - POS is a good cluster of words;
  - Smoothing is always helpful in statistics;


Thanks!